

Identifiability, regularity conditions and a two-metric adaptive projection for ℓ_1 -norm minimization

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July 3, 2025

Introduction and Outline

What is identifiability?

- Three examples of identifiability in optimization
- Identifiability in the language of slope

Regularity conditions and there relationship

- Slope Error Bound, PL, Quadratic Growth on (X, d) and $(\mathcal{M}, d_{Rie})$

A two-metric adaptive projection method

- Connections to two-metric projection for bound-constrained optimization
- Convergence properties under EB
- Numerical experiments

Summary and future work

Example 1: active sets in bound-constrained optimization

For a c^2 -smooth and strongly convex function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the optimality condition of the bound-constrained problem

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{subject to} \quad x^i \geq 0, i = 1, \dots, n.$$

is given by

$$0 \in \nabla f(x^*) + N_{\mathcal{X}}(x^*),$$

i.e.

$$\nabla f(x^*)^i \geq 0, \quad i = 1, \dots, n, \quad \text{and} \quad x^{*i} > 0 \Rightarrow \nabla f(x^*)^i = 0.$$

Projected gradient iteration

$$x_{k+1} = P(x_k - \alpha_k \nabla f(x_k)),$$

converge to the optimal solution x^* with

$$\text{dist}(0, \nabla f(x_k) + N_{\mathcal{X}}(x_k)) \rightarrow 0$$

Moreover, if

$$0 \in \text{ri}(\nabla f(x^*) + N_{\mathcal{X}}(x^*)),$$

iterates identify an active manifold: eventually

$$x_k \in \mathcal{M} \triangleq \{x \geq 0 : x_i = 0, \forall i \in \mathcal{A}(x^*)\}$$

where $\mathcal{A}(x^*) = \{i : x_i^* = 0\}$.

Example 2: sparsity in ℓ_1 -norm minimization

Consider ℓ_1 norm regularization problem, where f is a c^2 -smooth and strongly convex function

$$\min f(x) + \gamma \|x\|_1$$

the optimality condition is given by

$$0 \in \nabla f(x^*) + \gamma \partial \|x^*\|_1.$$

Proximal gradient iteration

$$x_{k+1} = \text{prox}_{\gamma \|\cdot\|_1}(x_k - \alpha_k \nabla f(x_k)),$$

converge to the optimal solution x^* with

$$\text{dist}(0, \nabla f(x_k) + \gamma \partial \|x_k\|_1) \rightarrow 0.$$

Let $\mathcal{M} \triangleq \{x : x_i = 0, \forall i \in \mathcal{A}(x^*)\}$, where $\mathcal{A}(x^*) = \{i : x_i^* = 0\}$. Then $\mathcal{M}$ is identifiable at x^* if $0 \in \nabla f(\bar{x}) + \gamma \text{ri } \partial \|\bar{x}\|_1$.

Example 3: matrix rank in SDP

For a c^2 -smooth and strongly convex function f , the optimality condition of the semidefinite programming problem

$$\min_{X \in \mathbb{S}^n} f(X) \quad \text{subject to} \quad X \succeq 0.$$

is given by

$$0 \in \nabla f(X^*) + N_{\mathbb{S}_+^n}(X^*).$$

Projected gradient iteration

$$X_{k+1} = P(X_k - \alpha_k \nabla f(X_k)),$$

converge to the optimal solution X^* with

$$\text{dist}(0, \nabla f(X_k) + N_{\mathbb{S}_+^n}(X_k)) \rightarrow 0.$$

Moreover, if

$$0 \in \text{ri}(\nabla f(X^*) + N_{\mathbb{S}_+^n}(X^*)),$$

iterates identify an active manifold: eventually

$$X_k \in \mathcal{M} \triangleq \{X : \text{rank}(X) = \text{rank}(X^*), X \succeq 0\},$$

Identifiability

Each example involves an active manifold of solutions and can be identified by diverse algorithms. Which means that

$$\text{dist}(0, \partial f(x_k)) \rightarrow 0 \implies x_k \in \mathcal{M}, \forall k \quad \text{large enough}$$

Hence high-dimensional nonsmooth optimization

$$\min f(x)$$

reduce locally to low-dimensional smooth optimization on $\mathcal{M}$. This makes Newton-type method possible and avoid solving high dimensional Newton system.

Some basic properties of slope

More general, in the language of slope, identifiability can be rewrite as

$$|\nabla f|(x_k) \rightarrow 0 \implies x_k \in \mathcal{M}, \forall k \quad \text{large enough}$$

slope

The slope of a function f at a point x is defined as: if x is not a local minimizer

$$|\nabla f|(x) = \limsup_{y \neq x, y \rightarrow x} \frac{f(x) - f(y)}{d(x, y)}.$$

otherwise, $|\nabla f|(x) = 0$.

Properties of slope

- $|\nabla f|(x) \geq 0$ for all x , $|\nabla f|(x) = 0$ if x is a local minimizer.
- If f is convex at x , then $|\nabla f|(x) = \text{dist}(0, \partial f(x))$.
- If f is differentiable at x , then $|\nabla f|(x) = \|\nabla f(x)\|$.

Definition of identifiability on (X, d)

Definition

Given a function f , a critical sequence for a point $\bar{x} \in \text{dom } f$ is a sequence $x_k \rightarrow_f \bar{x}$ such that $|\nabla f|(x_k) \rightarrow 0$, and $\bar{x}$ is critical if such a sequence exists.

Definition

For a function f on a metric space X , consider a point $\bar{x} \in \text{dom } f$, and a set $\mathcal{M} \subset X$.

- 1 $\mathcal{M}$ is identifiable at $\bar{x}$ if its complement contains no critical sequence for $\bar{x}$.
- 2 The modulus of identifiability for $\mathcal{M}$ at $\bar{x}$ is

$$\liminf_{x \rightarrow_f \bar{x}, x \notin \mathcal{M}} |\nabla f|(x)$$

Remark

$\mathcal{M}$ is identifiable at $\bar{x}$ iff the modulus of identifiability is strictly positive. Moreover, if $\bar{x}$ is not critical, then every set containing $\bar{x}$ is identifiable.

Proposition

For a function f on a metric space X , consider a critical point $\bar{x} \in \text{dom } f$, and a set $\mathcal{M}$ such that $x \in \mathcal{M} \subset X$ is identifiable at $\bar{x}$. If $x_k \rightarrow_f \bar{x}$, then

$$|\nabla f|(x_k) \rightarrow 0 \implies x_k \in \mathcal{M}, \forall k \quad \text{large enough}$$

Uniqueness

Definition

$\mathcal{M}$ is a strongly critical identifiable sets at $\bar{x}$ if all sequences $\{x_k\}$ in $\mathcal{M}$ satisfying $x_k \rightarrow_f \bar{x}$ are critical for $\bar{x}$. i.e. $|\nabla f|(x_k) \rightarrow 0$

Under this definition, the identifiable set is unique locally.

Proposition (Uniqueness of strongly critical identifiable sets)

Given a function f on a metric space, for any two strongly critical identifiable sets $\mathcal{M}_1$ and $\mathcal{M}_2$ at a point $\bar{x}$, there exists an f -attentive neighborhood $B_\epsilon^f(\bar{x})$ with $\epsilon > 0$ such that

$$\mathcal{M}_1 \cap B_\epsilon^f(\bar{x}) = \mathcal{M}_2 \cap B_\epsilon^f(\bar{x})$$

Identifiability in Euclidean space

Definition

Given a closed function $f: \mathbb{R}^n \rightarrow (-\infty, +\infty]$, consider a point $\bar{x} \in \text{dom } f$ and a manifold $\mathcal{M}$ containing $\bar{x}$. Then f is partly smooth at $\bar{x}$ for a subgradient $\bar{y} \in \partial f(\bar{x})$ relative to $\mathcal{M}$ if it satisfies the following properties:

Prox-regularity: the function f is prox-regular at $\bar{x}$ for $\bar{y}$.

Restricted smoothness: the restriction $f|_{\mathcal{M}}$ is smooth around $\bar{x}$.

Sharpness: the subspace $\text{par } \hat{\partial} f(\bar{x})$ is just the normal space $N_{\mathcal{M}}(\bar{x})$.

Inner semicontinuity: for all $y \in \partial f(\bar{x})$ near $\bar{y}$ and sequences $x_r \rightarrow \bar{x}$ in $\mathcal{M}$, there exist $y_r \in \partial f(x_r)$ converging to y .

Theorem (Identifiability and partial smoothness)

Consider a closed function $f: \mathbb{R}^n \rightarrow (-\infty, +\infty]$ with slope zero at a point $\bar{x}$ lying in a manifold $\mathcal{M}$. Suppose that $f|_{\mathcal{M}}$ is smooth around $\bar{x}$. Then $\mathcal{M}$ is identifiable at $\bar{x}$ if and only if f is partly smooth at $\bar{x}$ for zero relative to $\mathcal{M}$ with $0 \in \text{ri } \hat{\partial} f(\bar{x})$.

Uniqueness in Euclidean space

Proposition

If a closed function $f : \mathbb{R}^n \rightarrow (-\infty, +\infty]$ is partly smooth for zero at a point $\bar{x}$ relative to a manifold $\mathcal{M}$, then $\mathcal{M}$ is strongly critical at $\bar{x}$.

Remark

identifiable manifolds in Euclidean space are strongly critical identifiable sets therefore unique locally.

Regularity conditions

slope Error-Bound

We say f satisfies the slope Error Bound with constant $\mu > 0$ (μ - EB) around $\bar{x}$ if

$$\mu \operatorname{dist}(x, S) \leq |\nabla f|(x)$$

hold for all x in some neighborhood of $\bar{x}$. Where S is an arbitrary set containing $\bar{x}$.

PL

$$(f(x) - f^*) \leq \frac{1}{2\mu} |\nabla f|^2(x)$$

Quadratic Growth

$$\frac{\mu}{2} \operatorname{dist}^2(x, S) \leq f(x) - f^*$$

Relationship between regularity conditions

[Drusvyatskiy et al., 2021][Theorem 3.7] states that

$$(X, d) \quad \text{PL (KL)} \implies \text{slope EB}$$

and

$$(\mathbb{R}^n, d) \quad \text{slope EB} \iff \text{subdifferential EB}$$

subdifferential EB

We say f satisfies the subdifferential Error Bound with constant $\mu > 0$ ($\mu - \text{EB}$) around $\bar{x}$ if

$$\mu \text{dist}(x, S) \leq \text{dist}(0, \hat{\partial}f(x))$$

hold for all x in some neighborhood of $\bar{x}$. Where S is an arbitrary set containing $\bar{x}$.

[Drusvyatskiy et al., 2013][Corollary 3.2] states that

metric subregularity of $\partial f \implies$ Quadratic Growth

where $S = (\partial f)^{-1}(0)$

relationship on (X, d)

PL (KL) $\implies$ slope EB $\implies$ Quadratic Growth

Now suppose $f|_{\mathcal{M}}$ is C^2 -smooth around $\bar{x}$, $\mathcal{M}$ is a C^2 -smooth identifiable manifold with Riemannian metric d_{Rie} , then [Rebjock and Boumal, 2024] states that

relationship on $(\mathcal{M}, d_{Rie})$

PL (KL) $\iff$ slope EB $\iff$ Quadratic Growth

Note that these conditions are limited to the manifold $\mathcal{M}$, and

$$|\nabla f|(x) = \|\text{grad}_{\mathcal{M}} f(x)\|$$

relationship between (X, d) and $(\mathcal{M}, d_{Rie})$

[Lewis and Tian, 2024][Theorem 5.12, Proposition 8.6] states that

$$\text{PL (KL) on } (X, d) \iff \text{PL (KL) on } (\mathcal{M}, d_{Rie})$$

However, there is a critical flaw in the proof $\implies$. Actually, follow the idea of [Lewis and Tian, 2024], we can also prove that

$$\text{slope EB on } (X, d) \iff \text{slope EB on } (\mathcal{M}, d_{Rie})$$

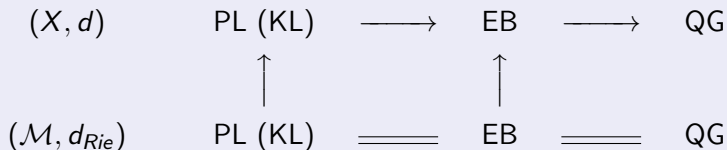
Theorem (Quadratic growth)

On a complete metric space, suppose that a closed function f has slope zero at a point $\bar{x}$, and consider any identifiable set $\mathcal{M}$ at $\bar{x}$. Then f has quadratic growth around $\bar{x}$ if and only if it has quadratic growth around $\bar{x}$ relative to $\mathcal{M}$. Indeed, the two growth rates are identical:

$$\liminf_{\substack{x \rightarrow \bar{x} \\ x \neq \bar{x}}} \frac{f(x) - f(\bar{x})}{d(x, \bar{x})^2} = \liminf_{\substack{x \rightarrow \bar{x}, x \in \mathcal{M} \\ x \neq \bar{x}}} \frac{f(x) - f(\bar{x})}{d(x, \bar{x})^2}.$$

Suppose that $f|_{\mathcal{M}}$ is C^2 -smooth around $\bar{x}$, $\mathcal{M}$ is a C^2 -smooth identifiable manifold with Riemannian metric d_{Rie} , then

conclusion



open question

What is the most general condition can make the following implication hold with $\mathcal{M}$ is a C^2 -smooth identifiable manifold with Riemannian metric d_{Rie} ?

$$\text{PL (KL) on } (X, d) \implies \text{PL (KL) on } (\mathcal{M}, d_{Rie})$$

In which cases EB holds?

convex composite optimization

$$\min_{x \in \mathbb{R}^n} f(x) + h(x)$$

where f is a smooth convex function and h is a proper closed convex function.

[Drusvyatskiy and Lewis, 2018][Theorem 3.3, Corollary 3.6] states that proximal EB is equivalent to subdifferential EB.

proximal-EB

For an optimal point $x^* \in X^*$, there exists a neighborhood $B(x^*, \delta)$ such that for all $x \in B(x^*, \delta)$,

$$\kappa r(x) \geq \text{dist}(x, X^*)$$

where $r(x)$ is a residual function $\|x - \text{prox}_h(x - \nabla f(x))\|$ and $\kappa > 0$ is a constant.

[Zhou and So, 2017] states that proximal EB holds for a class of structured convex optimization problems, including

- f is strongly convex, ∇f is Lipschitz continuous, h is arbitrary proper closed convex function.
- $f(x) = g(Ax)$, where g is a proper convex function satisfies
 - ▶ g is continuously differentiable on $\text{dom}(g)$.
 - ▶ g is strongly convex, ∇g is Lipschitz continuous on any compact subset of $\text{dom}(g)$.

and A is a linear operator. h is group LASSO regularizer.

- $f(x) = g(Ax) + \langle c, x \rangle$, where g is a proper convex function are as above. h has a polyhedral epigraph.

Bound-Constrained Problem

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{subject to} \quad x^i \geq 0, i = 1, \dots, n. \quad (1)$$

- $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is bounded below by f_{low} on the feasible region.

Two-Metric Projection [Bertsekas, 1982]

$$x_{k+1} := P(x_k - \alpha_k D_k \nabla f(x_k)),$$

- $P(z)$ is the projection onto the feasible region, i.e.

$$[P(z)]^i = \max\{z^i, 0\},$$

- $D_k \in \mathbb{R}^{n \times n}$, positive definite matrix
- To ensure descent in f , require $D_k[i, j] = 0, \forall i, j \in I_k^+, j \neq i$.

$$D_k = \left(\begin{array}{c|ccc} \bar{D}_k & 0 & \cdots & 0 \\ \hline 0 & d^{r_k+1} & & 0 \\ \vdots & & \ddots & \\ 0 & 0 & & d^n \end{array} \right)$$

$\underbrace{\hspace{10em}}_{I_k^+}$

$$I_k^+ \triangleq \{i \mid 0 \leq x_k^i \leq \epsilon_k, \nabla_i f(x_k) > 0\}.$$

Convergence properties

Main result

- Global convergence under suitable line search.
-

$$[\bar{D}_k^{-1}]_{ij} = \frac{\partial^2 f(x_k)}{\partial x^i \partial x^j} \quad \forall i, j \notin I_k^+.$$

Strict Complementarity + Local Strong Convexity $\Rightarrow$ Identification of Active Set ($I_k^+ = \mathcal{A}(x^*)$) + Quadratic Convergence rate.

- Eigenvalues of D_k should be uniformly bounded.
- Newton's equation should be solved exactly.

These requirements are rarely met in practice.

Bound-constrained formulation of ℓ_1 -norm minimization

ℓ_1 -norm regularization

$$\min_{x \in \mathbb{R}^n} \psi(x) = f(x) + h(x), \quad h(x) = \gamma \|x\|_1 \quad (2)$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a C^2 -smooth and convex function, and $\gamma > 0$ is a regularization parameter.

Let $x = x^+ - x^-$ in (2), where $x^+ = \max(0, x)$ and $x^- = -\min(0, x)$. Then (2) can be reformulated as the following constrained problem

Bound-constrained formulation

$$\begin{aligned} \min_{x^+, x^- \in \mathbb{R}^n} \quad & f(x^+ - x^-) + \gamma \sum_i [x_i^+ + x_i^-] \\ \text{s.t.} \quad & x^+ \geq 0, \quad x^- \geq 0. \end{aligned} \quad (3)$$

Why Two-metric projection can not be used directly?

Given an ϵ_k , let I_k^1 and I_k^2 are the “ I_k^- part” corresponding to x_k^+ and x_k^- respectively. Then we have

$$I_k^1 = \{i \mid 0 \leq (x_k^+)^i \leq \epsilon_k, g_k^i + \gamma \leq 0\} \cup \{i \mid (x_k^+)^i > \epsilon_k\}$$

$$I_k^2 = \{i \mid 0 \leq (x_k^-)^i \leq \epsilon_k, -g_k^i + \gamma \leq 0\} \cup \{i \mid (x_k^-)^i > \epsilon_k\}$$

- $I_k^1 \cap I_k^2 \neq \emptyset$ if $(x_k^+)^i > \epsilon_k, (x_k^-)^i \leq \epsilon_k, -g_k^i + \gamma \leq 0$ or $(x_k^-)^i > \epsilon_k, (x_k^+)^i \leq \epsilon_k, g_k^i + \gamma \leq 0$.
- $\bar{D}_k$ is singular if $I_k^1 \cap I_k^2 \neq \emptyset$.
- Newton's equation is unsolvable even if $\nabla^2 f \succ 0$.
- Inexact approximation $\nabla^2 f(x_k) + \mu_k I$ will lead to numerical instability when μ_k is small.

Two-metric Adaptive Projection

- Slightly modify the definition of I_k^1 and I_k^2 and reduce the problem size by aggregating x_k^+ and x_k^- to avoid the intersection.

$$I_k^{-+} \triangleq \left\{ i : \begin{array}{l} x_k^i > \epsilon_k \text{ or} \\ 0 \leq x_k^i \leq \epsilon_k, g_k^i \leq -\gamma \end{array} \right\} \subseteq I_k^1,$$

$$I_k^{--} \triangleq \left\{ i : \begin{array}{l} x_k^i < -\epsilon_k \text{ or} \\ -\epsilon_k \leq x_k^i \leq 0, g_k^i \geq \gamma \end{array} \right\} \subseteq I_k^2,$$

$$I_k^- \triangleq I_k^{-+} \cup I_k^{--}, I_k^{-+} \cap I_k^{--} = \emptyset.$$

- The Newton's equation doesn't need to be solved exactly i.e.

$$(H_k + \mu_k I)[p_k]_{I_k^-} = [g_k + \omega_{k,\epsilon}]_{I_k^-} + r_k$$

- Reserve **identification of active set** ($I_k^+ = \mathcal{A}(x^*)$) and **superlinear convergence** under a weaker condition than local strong convexity.

Two-metric Adaptive Projection

$$x_{k+1} := \mathcal{P}_{k,\epsilon}(x_k - t_k p_k), \quad \forall k \geq 0,$$

where $t_k > 0$ is the stepsize determined by line search. The step p_k is defined as below:

$$[p_k]_{I_k^+} \triangleq [g_k + \omega_{k,\epsilon}]_{I_k^+}$$

and $[p_k]_{I_k^-}$ satisfies

$$(H_k + \mu_k I)[p_k]_{I_k^-} = [g_k + \omega_{k,\epsilon}]_{I_k^-} + r_k \quad (4)$$

where H_k is a symmetric positive semi-definite matrix. μ_k is a positive scalar such that

$$\mu_k = c \left\| \begin{bmatrix} [x_k - \text{Prox}_h(x_k - g_k)]_{I_k^+} \\ [g_k + \omega_k]_{I_k^-} \end{bmatrix} \right\|^\delta, \quad \delta \in (0, 1),$$

and $r_k \in \mathbb{R}^{|I_k^-|}$ is a residual that satisfies the following condition for a fixed $\tau \in (0, 1)$:

$$\|r_k\| \leq \tau \min\{\mu_k \| [p_k]_{I_k^-} \|, \| [g_k + \omega_k]_{I_k^-} \|\}$$

Two-metric Adaptive Projection

$\mathcal{P}_{k,\epsilon}$ and $\omega_{k,\epsilon}$ are associated to x_k, ϵ and defined as below (recall that S_t denotes the soft thresholding operator):

$$\mathcal{P}_{k,\epsilon}^i(v) = \begin{cases} \max\{v^i, 0\} & \text{if } i \in I_k^{-+} \\ \min\{v^i, 0\} & \text{if } i \in I_k^{--} \\ S_{t_k\gamma}(v^i) & \text{if } i \in I_k^{+} \end{cases}$$

$$\omega_{k,\epsilon}^i = \begin{cases} \gamma & \text{if } i \in I_k^{-+} \\ -\gamma & \text{if } i \in I_k^{--} \\ 0 & \text{if } i \in I_k^{+} \end{cases}.$$

Convergence properties

Hold for a class of structured convex optimization problems.

Main result

Strict Complementarity + Error Bound $\Rightarrow$ Identification of Active Set ($I_k^+ = \mathcal{A}(x^*)$) + Superlinear convergence $(1 + \delta)$.

Numerical experiments

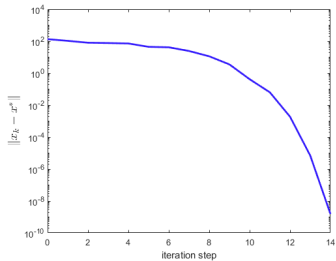
logistic regression

$$\min_{x \in \mathbb{R}^n} \frac{1}{m} \sum_{i=1}^m \log \left(1 + \exp \left(-b_i \cdot a_i^T x \right) \right) + \gamma \|x\|_1, \gamma = \frac{1}{m}$$

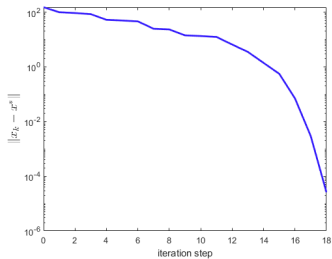
- Second-order method: IRPN [Yue et al., 2019] (SQA-type method), newGLMNET
- First-order method: SpaRSA, FISTA

datasets		IRPN	TMAP
rcv1_train	outer iter.	6	-
	inner iter.	492	13
	time	1.46	0.17
news20	outer iter.	9	-
	inner iter.	1197	18
	time	40.42	1.64
real-sim	outer iter.	13	-
	inner iter.	547	21
	time	4.30	1.11

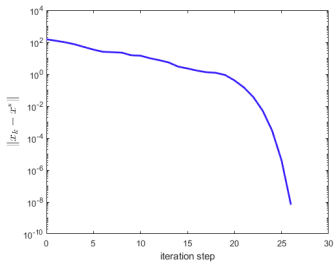
TMAP $\gg$ SQA-type method and newGLMNET (3-10 times associate to the sparsity) $>$ SpaRSA and FISTA.



rcv1



news20



real-sim

Numerical experiments

LASSO (large-scale reconstruction problem)

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} \|Ax - b\|_2^2 + \gamma \|x\|_1.$$

- Second-order method: ASSN [Xiao et al., 2018] (Semi-smooth Newton method).
- First-order method: SpaRSA, FPC_AS, ADMM

dataset size

$$n = 512^2 = 262144, m = n/8 = 32768, \text{ sparsity} = 2.5\%.$$

Table: N_A denotes the total number of calls to A and A^T and the CPU time (in seconds) is averaged over 10 independent runs.

Dynamic range		ASSN	TMAP
20dB	N_A	298.2	300.6
	time	1.31	1.30
40dB	N_A	459.2	408.9
	time	2.51	2.32
60dB	N_A	635.4	624.9
	time	2.29	2.23
80dB	N_A	858.2	791.5
	time	2.99	2.74

TMAP $\approx$ Semi-smooth Newton method $\gg$ SpaRSA, FPC_AS and ADMM.

Summary

	Our algorithm	Bertsekas algorithm
Problem class	ℓ_1 -norm minimization	Bound-constrained
Regularity condition	Error Bound	Local Strong Convexity
Newton's equation	Inexact	Exact
Global convergence	✓	✓
Local convergence rate	Superlinear $(1 + \delta)$	Quadratic

Competitive against the state-of-the-art algorithms for large-scale ℓ_1 -norm minimization (LASSO, logistic regression).

Future work

- Use BFGS to approximate the Hessian matrix in the Two-metric Adaptive Projection algorithm.
- Extend the algorithm to more general optimization problems.
- Identifiability requires the nondegeneracy/strict complementarity, in which case can this condition satisfied?
- In which cases the regularity conditions on (X, d) imply the regularity conditions on $(\mathcal{M}, d_{Rie})$? ($\mathcal{M}$ is identifiable manifolds)

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